AGI ALPHA FLYWHEEL

Executive brief | From a numerical failure to a verified digital answer

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A useful capability, ready to inspect

We delivered a runnable numerical repair package for a real public failure, then tested whether expert agent operation improved on competent ordinary use. All three workflows passed all 64 fresh checks. The useful engineering survives; an exclusive breakthrough or general Operational Alpha claim does not.

A scientific software recipient can reproduce an extreme-tail beta-quantile failure, obtain a checked scalar answer, inspect refusal and underflow behavior, and decide whether to integrate the component. In the installed SciPy 1.17.0 environment, one reported case returned **0.5** where the verified answer is approximately $2.844444444444444 \times 10^{-33}$. Several failed calls emitted warnings. The public issue supplied the failure and candidate formulation; we preserve that credit.

What the experiment established

Workflow	Fresh numerical/endpoints	Correct refusals	Recorded construction
Ordinary competent operation	56/56	8/8	10.16 min
Expert operation	56/56	8/8	13.66 min
Strong conventional workflow	56/56	8/8	8.22 min

The early conventional challenge solved all five public examples. All execution arms received the same report, recipe, libraries, permissions and contract. Ordinary use could plan, verify and recover. These are **64 cases on one public mission**, with one implementation trial per workflow. Model backend identity, token usage and provider bills were not independently exposed; this was live assistant/tool execution, with no simulated success or graphical-interface benchmark.

Independent audit verified all 98 frozen files and the freeze order, rescored acceptance, and passed 17 separate mathematical controls. Rebuilding source reproduced the result. Existing journal/accounting/fitness components were reused with unknown costs and zero receipts.

A critical boundary stays beside the headline. On an already-known midpoint diagnostic outside the fresh bank, ordinary and expert returned the correct smallest positive binary64 value; the original conventional arm incorrectly returned underflow zero. A separately labeled conventional correction passed the 64-case replay plus that diagnostic. This was not fresh confirmation and did not establish an expert advantage over ordinary use.

Three decisions

- **Commercial delivery:** the research contract accepts the usable digital artifact. Paid customer acceptance, demand and realized profit remain unobserved.
- **Operational Alpha:** not established. Expert operation did not improve fresh accepted quality over competent ordinary use; complete cost metering and multi-mission replication are missing.
- **Transformational contribution:** no exclusive numerical advance. Strong established methods deliver the outcome. The original formulation came from the public report.

The service must earn its price

All amounts below are nominal US dollars. The released artifact is a free outside option. Any paid offering must provide buyer-specific qualification, integration, handoff and bounded support. That additional engagement has not been delivered to a customer. Proposed prices and service effort are transparent assumptions, not market quotations.

Let M be unknown additional model/tool cost per future engagement, including retries. Let D be unknown historical cost of the whole research and publication project, charged once. Four hypothetical central engagements at **$1,500** yield:

Measure	Complete conditional central result
Contribution per engagement	$879.075 - M$
Cohort operating profit	$2,716.30 - 4M$
Full-project return	$2,216.30 - 4M - D$
Operating break-even model cost	$M < \$679.075$
At illustrative $M = \$100$	Operating profit $2,316.30; full-project return $1,816.30 - D$
Funding at illustrative $M = \$100$	**$3,966.25**, including a $1,875 refund/correction reserve
Accepted milestone alternative	**$3,476.55** funding; extra payment fees reduce return by $1.20

The $100 model-cost coordinate is a sensitivity, not a measured bill. Central operating break-even price is $890.45 at that coordinate. Favorable operating profit is $16,526.93 - 8M$; an adverse zero-demand launch loses $1,650 operating and $3,050 plus historical D after premature setup commitments. A conditional-commitment gate can avoid those prospective launch costs.

The central buyer-value ceiling is only **$1,650**: eight avoided qualification hours at $150/h plus $600 avoided delay, less one participation hour. At the proposed fee, modeled buyer surplus is $150. A buyer who can reuse the free package quickly may pay nothing. No customer interviews, commitments or collections were observed.

Funding is separate from profit. Available unrestricted funds are unknown. Earned milestone receipts can reduce the cash gap only after useful accepted work; refundable deposits remain restricted. The reserve covers one stated refund/correction stress, not unlimited correlated failures. The staged-payment model preserves obligations and includes both transaction fees. Conventional teams receive the same sharing rights and assumptions; ordinary sharing produces no exclusive expert interaction.

Recommendation and delivery

Retain the engineering and pursue only a concretely scoped, customer-funded qualification pilot. Stop the exclusive algorithmic-novelty claim. Recommend the ordinary implementation as the research entry point; preserve all three originals, the conventional repair, failed attempts, evaluator corrections and prior kinetics evidence. No fresh-transfer benefit or Gym investment is claimed.

The edition includes LaTeX, bibliography, vector figures, PDF/HTML, runnable source, frozen evidence, economics, claim map and a LinkedIn draft for approval only.

From the extracted package, rebuild and replay with:

```
python -B delivery/reproduce.py --output ../beta-replay --native
```

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